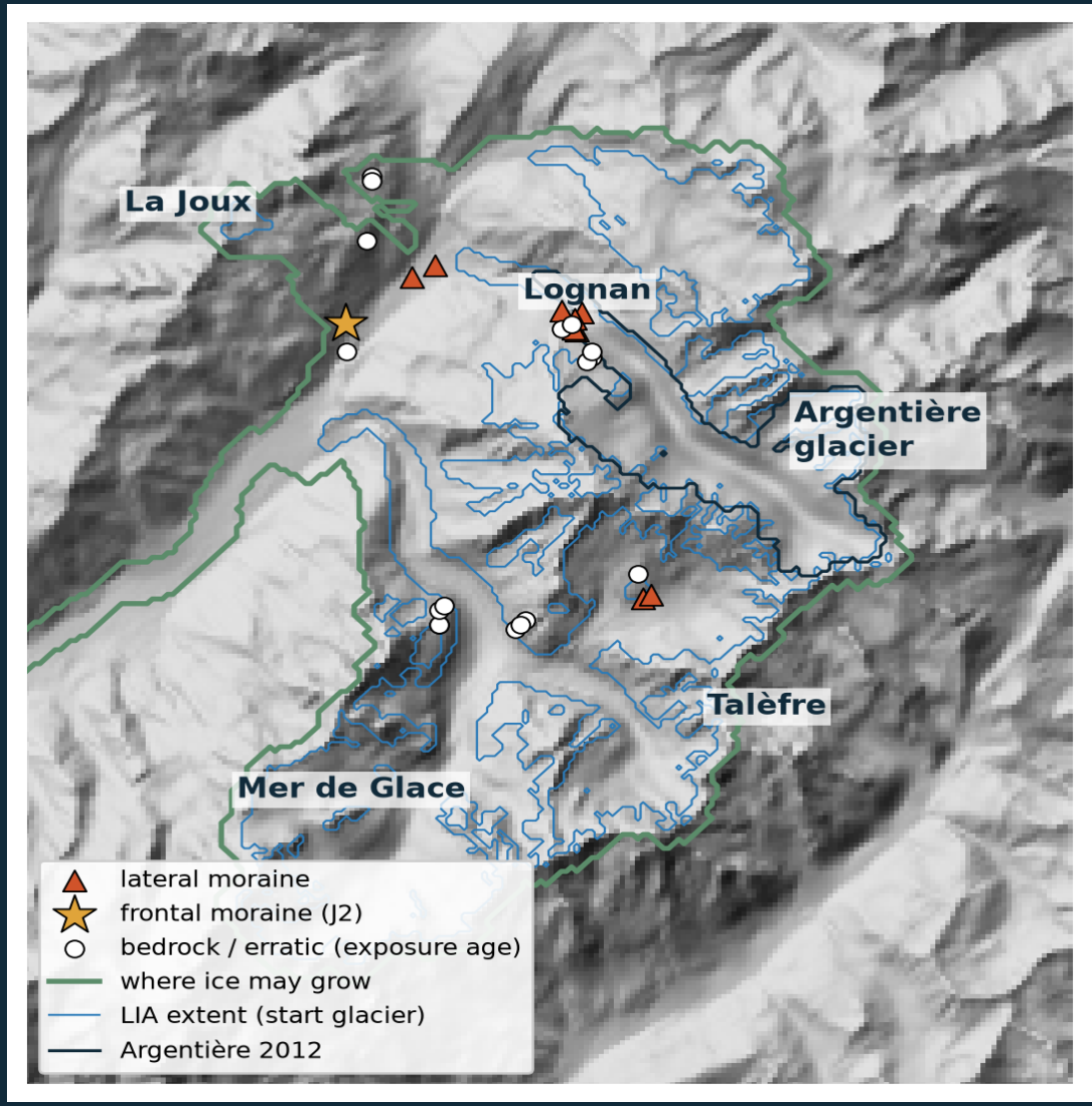
Younger Dryas temperature from the Argentière moraines

Inverting T(t), 13.6–9.0 ka, from dated landforms with a differentiable glacier model

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What made the Younger Dryas glacier?

Climate parameters (27)

T(t), precipitation P, seasonality T_s, melt factor DDF

Glacier parameters (2)

ice softness A (flow law) and basal sliding τ_ref

Goal: the values with which the glacier model reproduces the dated Younger Dryas landforms

Observations

dated moraines and exposure ages, 13.6–9 ka

27

landforms

Cheap first run · 200 m

Invert **T(t) only** to get the prior. The rest of parameters from the literature

35.3 → 25.9

loss

T(t) inversion · 100 m

starts from the 200 m T(t) (S 9–11). Then we do a sensitivity analysis (S 10)

→ 22.7

loss

Joint inversion · 100 m

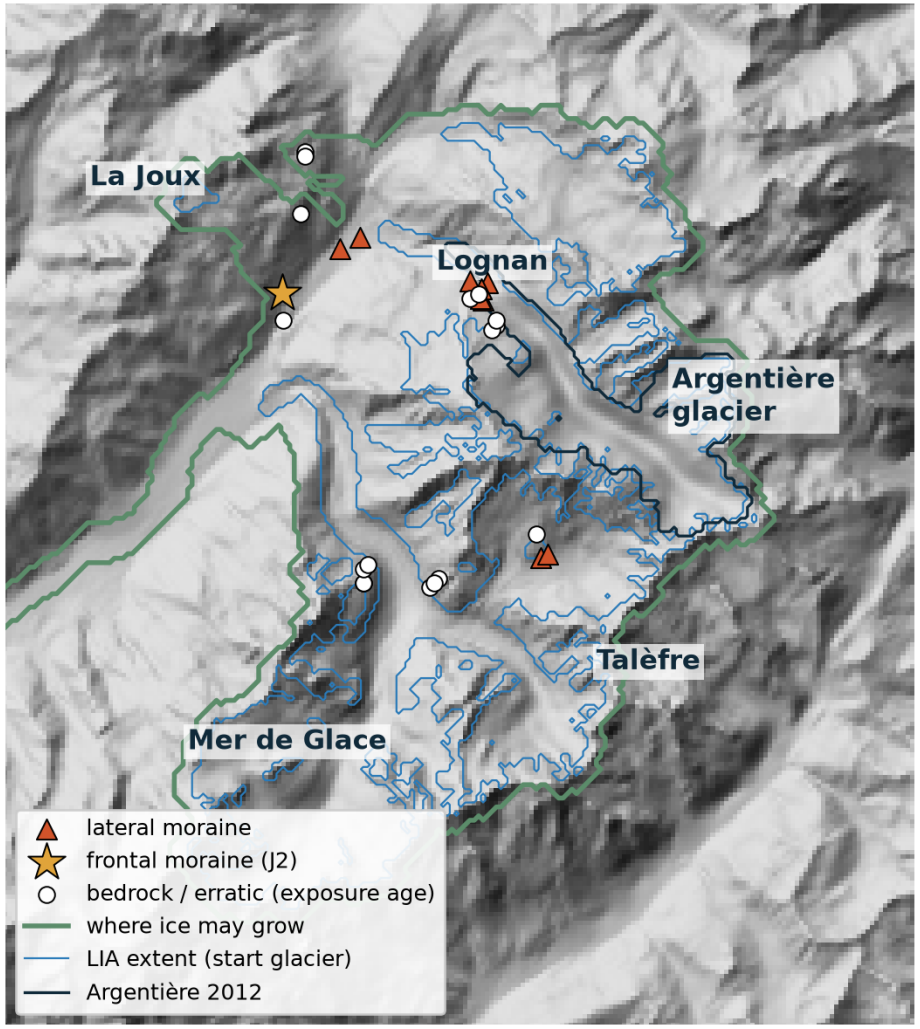
T(t) + P, T_s, DDF, A, τ_ref (slides 12–14)

→ 13.4

loss

*All results from here on are at 100 m spatial resolution; the 200 m run only provides the prior for **T(t)**.*

Dated landforms record the Lateglacial glacier



landform	n	age (ka)	constrains
Frontal moraine J2 (La Joux)	1	10.8	terminus position
Lateral, dated: Lognan L1–L3, Talèfre T1–T2	5	11.9–10.3	ice-surface height
Lateral, undated: La Joux, Lognan	4	13.1–10.8 *	ice-surface height
Bedrock / erratic exposure ages	17	19.1–9.8	ice-free after exposure

* age window from morphostratigraphy

^{10}Be ages: AlpIce-1 (Kamleitner et al. 2026); Protin et al. 2019, 2021; Lehmann et al. 2019, 2020

Start glacier: LIA extent and surface (Reinthaler & Paul 2025)

Grid: 100 m, 250 × 250 cells

A forward run: from T(t) to a glacier

T(t) at 2000 m
24 nodes, the unknown

→ **T(x, y, t)**
lapse rate

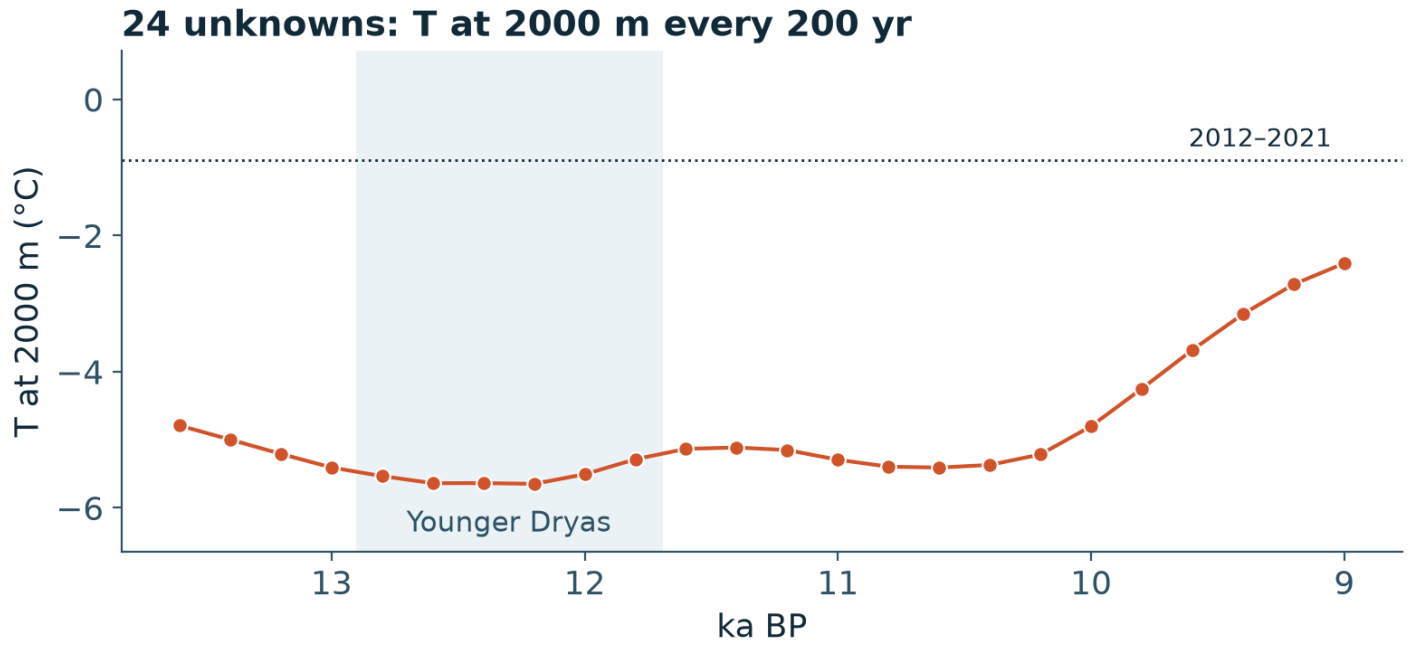
→ **Mass balance**
P, T_s, DDF, T_present

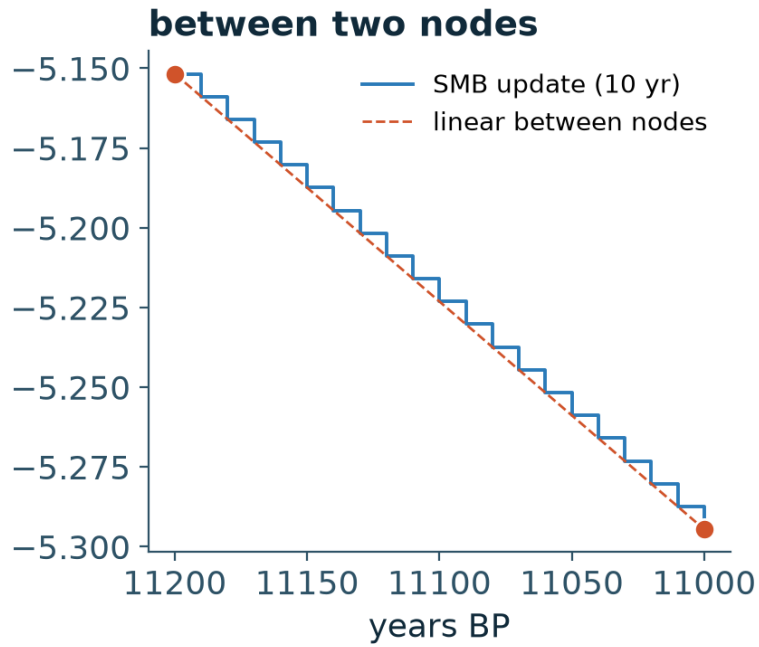
→ **Ice flow**
IGM emulator, fixed

→ **H(x, y, t)**
saved every 100 yr

→ **Loss**
vs dated landforms

= parameters borrowed from the literature (next slide), tested for sensitivity at the end



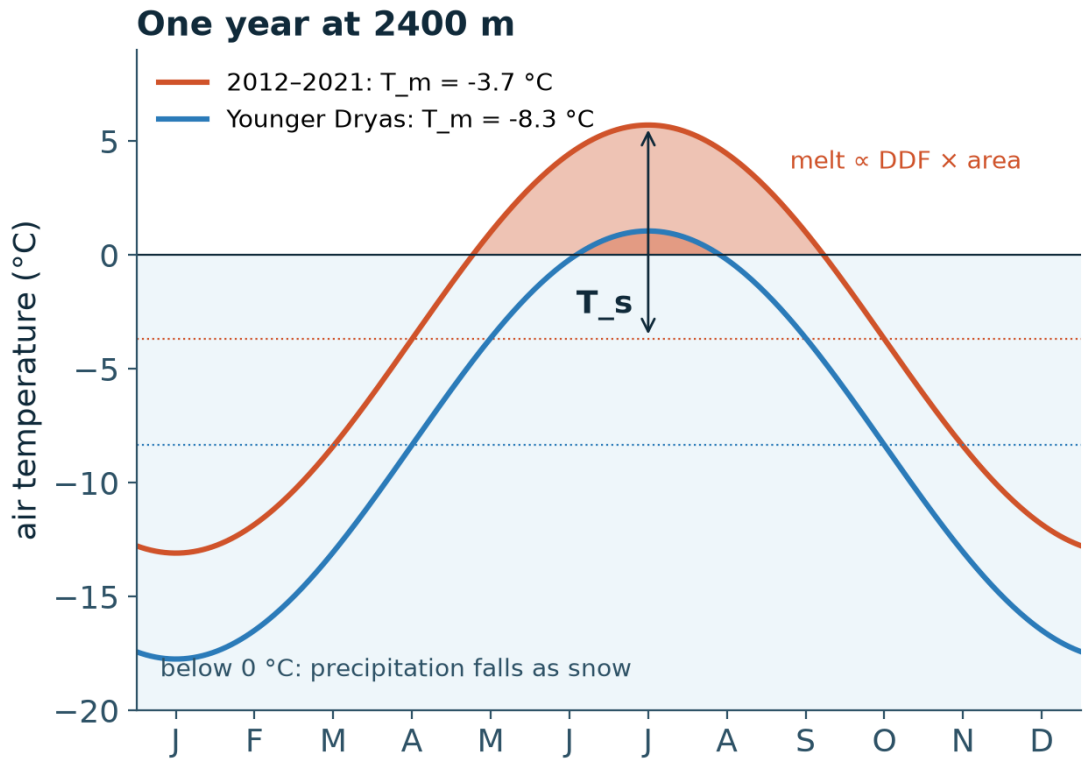


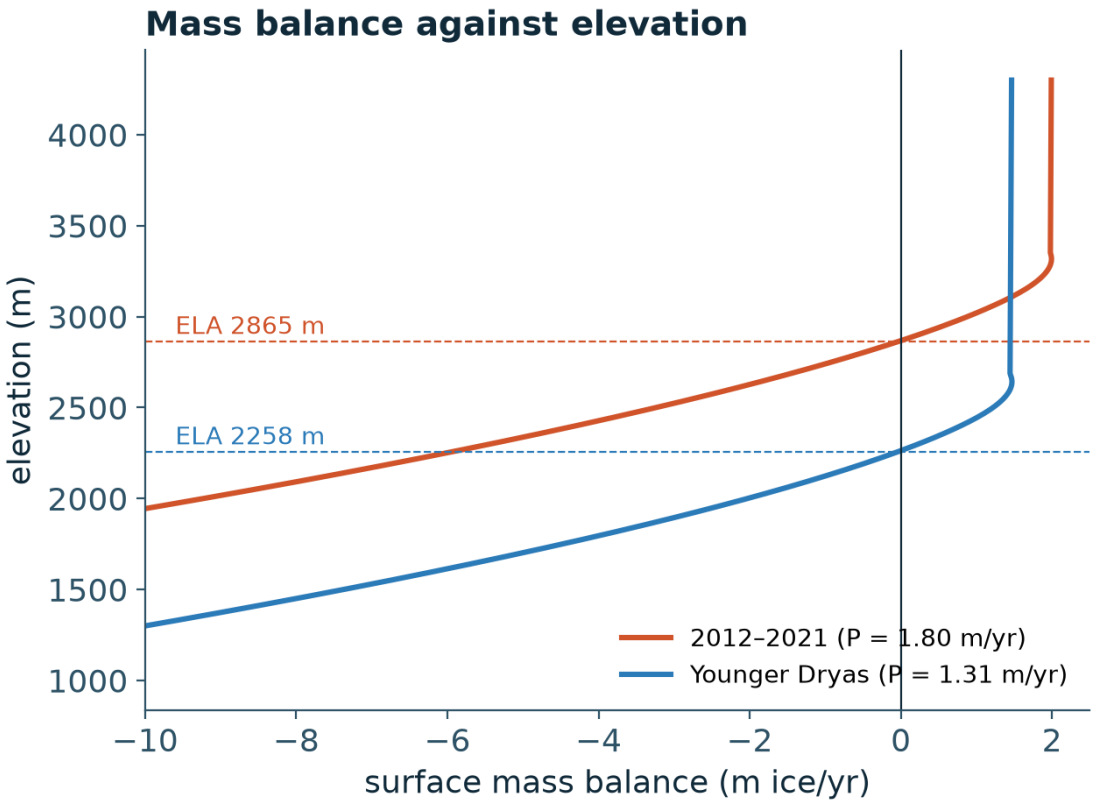
Fixed inputs, borrowed from the literature

parameter	value	what it does	source	tested range
P	1.8 m/yr	precipitation; snow where T < 0 °C	Chamonix 1.28 (1042 m), Col du Dôme ~2.5 (4250 m)	**1.5 – 2.1**
P scaling	+7 %/°C	Lateglacial P follows T(t)	Clausius–Clapeyron	×0.75 – ×1.25
T_s	9.4 °C	seasonal half-range of the year	Chamonix 1981–2010 (Jan −2.2, Jul 16.5 °C)	**7.9 – 10.9**
lapse rate	7 °C/km	T(z) from T at 2000 m	standard value	6 – 8
DDF	10 mm/K/d	melt per positive degree-day	upper literature bound	**8 – 12**
T_present	−0.89 °C	2012–2021 at 2000 m: anomaly reference	fitted to geodetic −0.66 m w.e./yr (Davaze 2020)	re-fitted
A (flow law)	78 MPa^{-3}/yr	ice softness	IGM	**52 – 117**
τ_ref (sliding)	0.215 MPa	basal sliding	IGM emulator	**0.15**
speed cap	600 m/yr	stability in the Lognan icefall	model choice	1000
T bounds	envelope	keeps T(t) in the literature range	Protin 2019, Heiri 2015, Samartin 2012	–

Check: P = 1.8 m/yr → 1.40 m w.e./yr accumulation vs 1.47 measured (Davaze et al. 2020)

How T(t), T_s and P make the mass balance





$$T_m(z, t) = T(t) - 7\ °C/km \cdot (z - 2000\ m)$$

T(t) shifts the whole field up or down

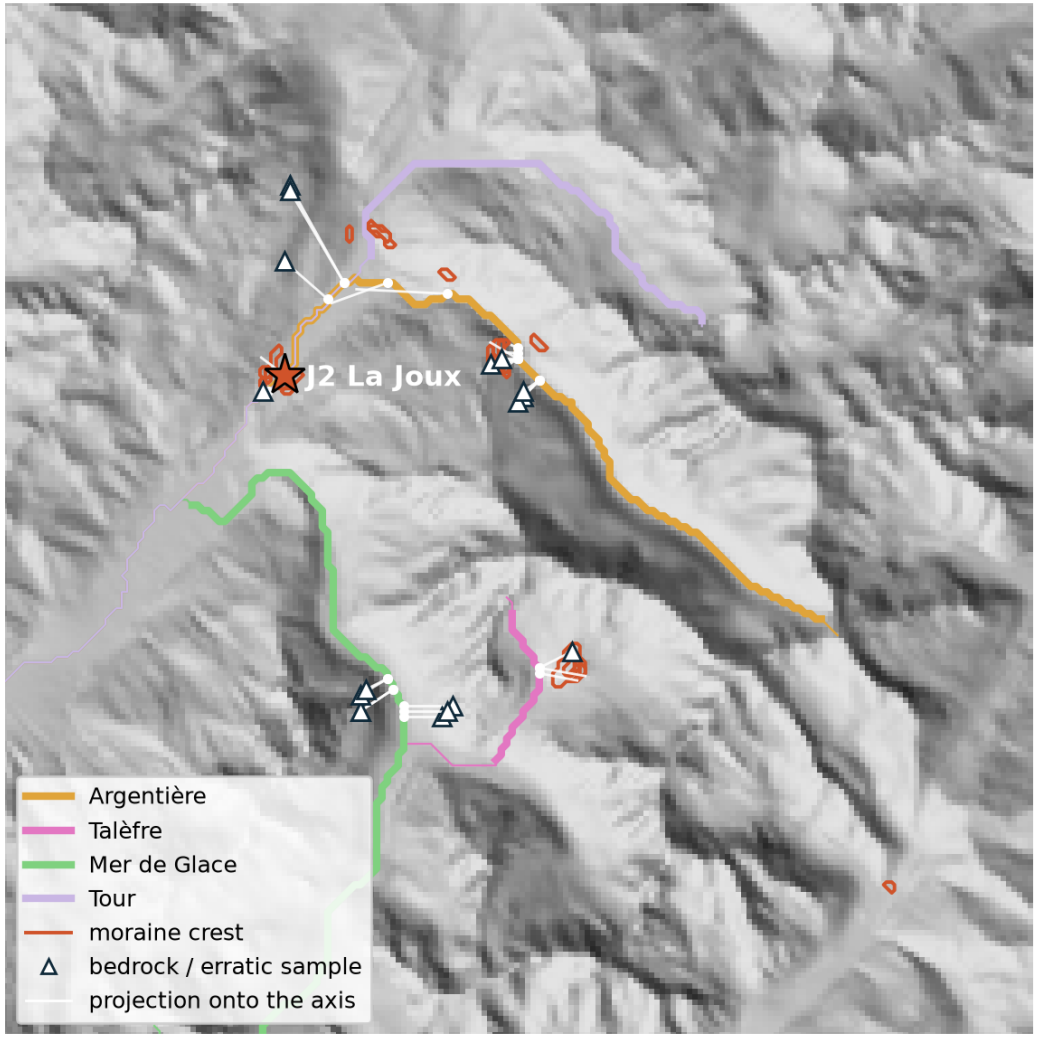
$$T(\tau) = T_m - T_s \cdot \cos 2\pi\tau$$

T_s sets how far summer rises above the mean

$$SMB = P(t) \cdot f(T < 0) - DDF \cdot \Sigma \max(T, 0)$$

P(t) = 1.8 m/yr · 1.07^(T(t) − T_present)

The loss: landforms read along their valley

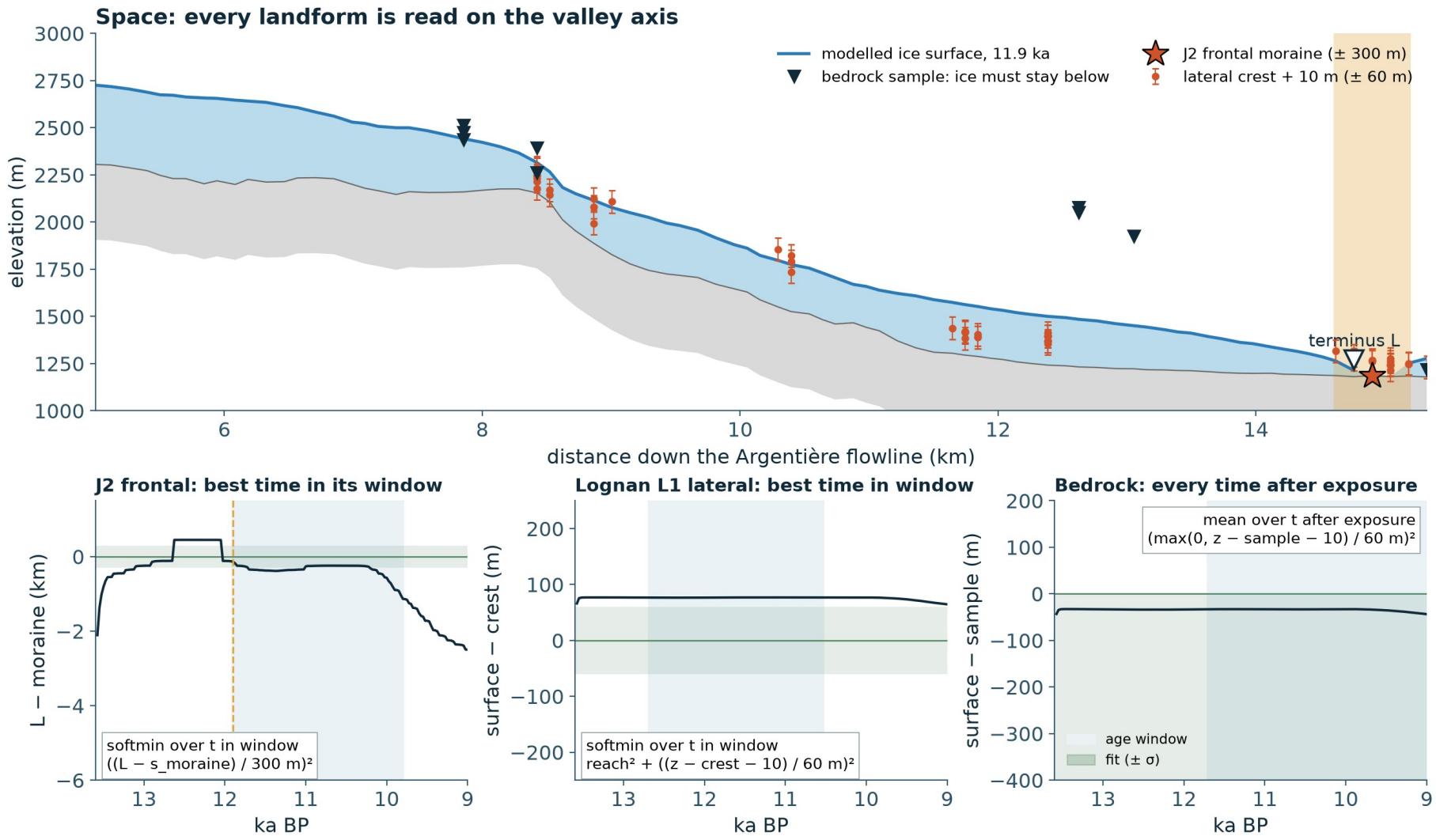


landform	asks	σ
frontal moraine	terminus L = moraine, at some time in its age window	300 m
lateral moraine	glacier reaches it, and axis surface = crest + 10 m, at some time in its window	300 m, 60 m
bedrock, erratic	axis surface ≤ sample + 10 m at every time after exposure	60 m
moraine preserved	no later advance overrides it	weight 0.5

$$J = \Sigma_{\text{landforms}}\ w \cdot \text{Huber}(\text{misfit}) \ + \ \Sigma\,(\Delta\Delta T)^2$$

- Terminus L: soft maximum of ice along the flowline — defined and differentiable even when the glacier stops short
- Each landform counts once; beyond 3σ it pulls linearly (Huber)
- Second term: smoothness of the 24 nodes

Scoring one forward run



Space

Each landform is moved to its cross-section of the valley axis.

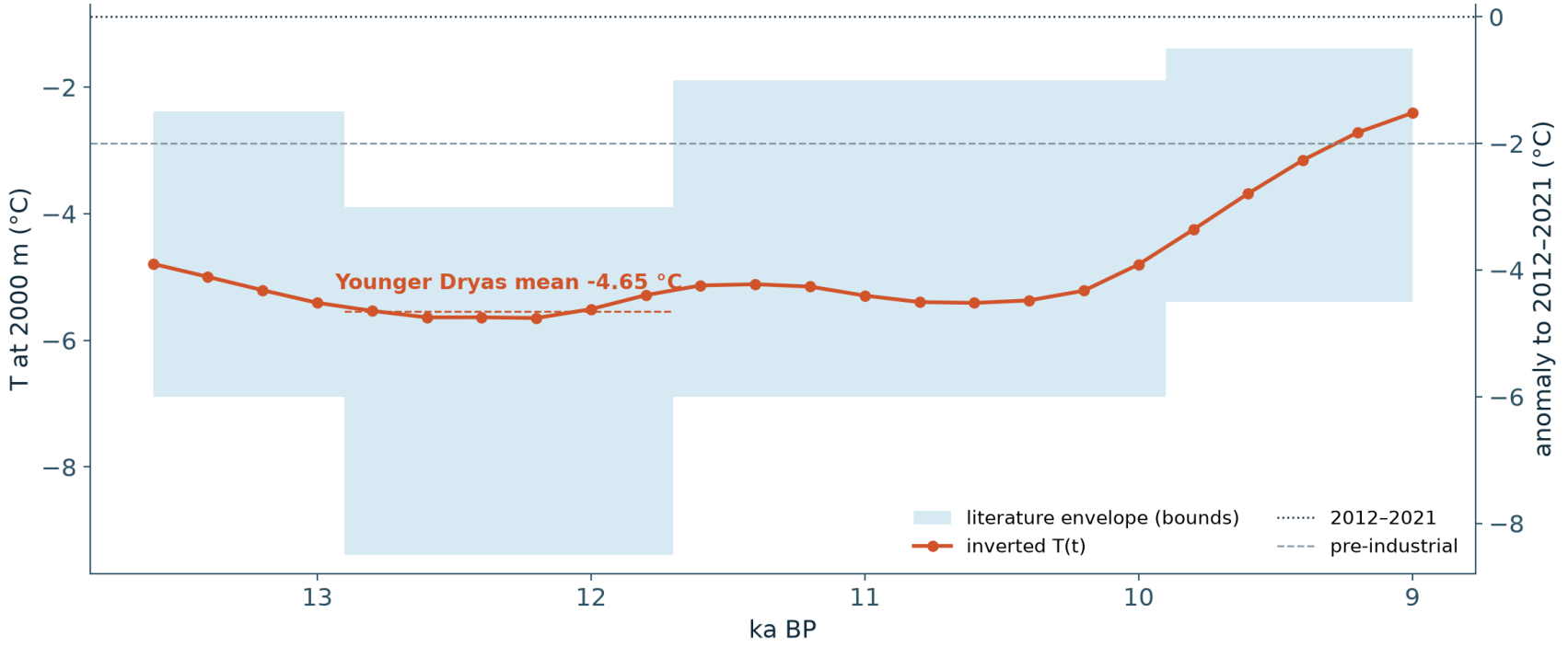
Time

Moraines: the best time in their age window. Bedrock: every time after exposure.

Score

(distance / σ)², summed over landforms.

Younger Dryas: 4.65 °C colder than 2012–2021



−4.65 °C

YD mean vs 2012–2021

−2.65 °C

YD mean vs pre-industrial

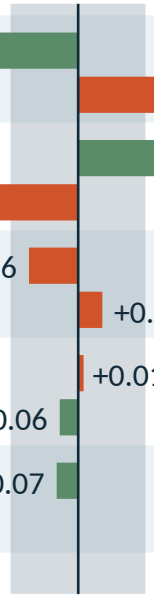
−5.65 °C

coldest node, 12.2 ka (at 2000 m)

check	value
Protin et al. 2019 (summer, YD / early Holocene)	3.0–5.5 °C below present
Loss over 30 iterations (from the 200 m result)	23.66 → 22.74
Nodes on the envelope bounds	0 of 24

Small temperature shifts, large changes in fit

parameter	low / high	change of the YD anomaly (°C)	loss
P	1.5 2.1 m/yr	−0.54 +0.47	19.3 25.5
DDF	8 12 mm/K/d	+0.34 −0.33	20.3 24.9
T_s	7.9 10.9 °C	−0.16 +0.08	23.5 22.4
A (flow law)	52 117	+0.01 −0.06	28.0 18.3
τ_ref (sliding)	0.15 MPa	−0.07	13.7



upper: low end · lower: high end better than base (22.4) worse data ±0.22 °C

Forward runs, best uniform T shift relative to the base parameters; mass-balance cases re-fit T_present.

Loss at the best T shift

22.4
literature parameters

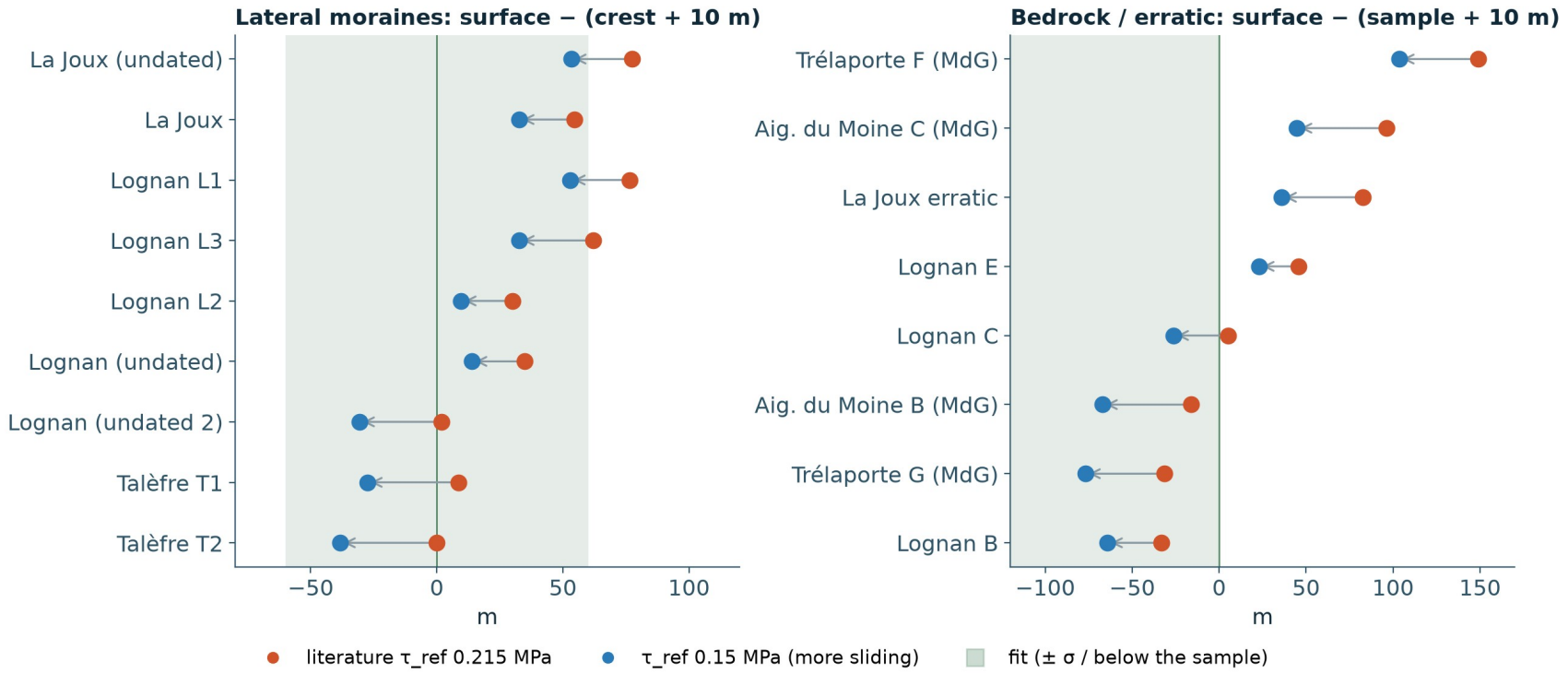
13.7
τ_ref 0.15 MPa (more sliding)

YD moves ≤ 0.54 °C; the fit moves far more

Still to come

- Forward checks: Lateglacial P, lapse rate
- First-order gradients were unreliable here (τ_ref: −0.80 predicted)
- **Joint inversion T(t) + P, T_s, DDF, A, τ_ref: slides 12–14**

Best fit: more sliding, same temperature



−36 %

loss, without changing T(t)

20–50 m

thinner ice at the landforms, same length

−0.07 °C

YD anomaly at the new best shift

same T(t)	τ_ref 0.215 (literature)	τ_ref 0.15
loss	22.7	14.5
landforms with misfit > 1	10	6
J2 terminus vs moraine	−129 m	−122 m

Softer ice (A 117) points the same way: loss 18.3

Why free the parameters too

Glacier too thick for its length

Right length (J2 met), but ice 55–80 m above the La Joux and Lognan crests. T(t) moves length and thickness together.

Ice flow thins it, T barely moves

τ_ref 0.15 MPa: loss 22.4 → 13.7, YD anomaly −0.07 °C. Softer ice the same way.

Invert T(t) + five parameters

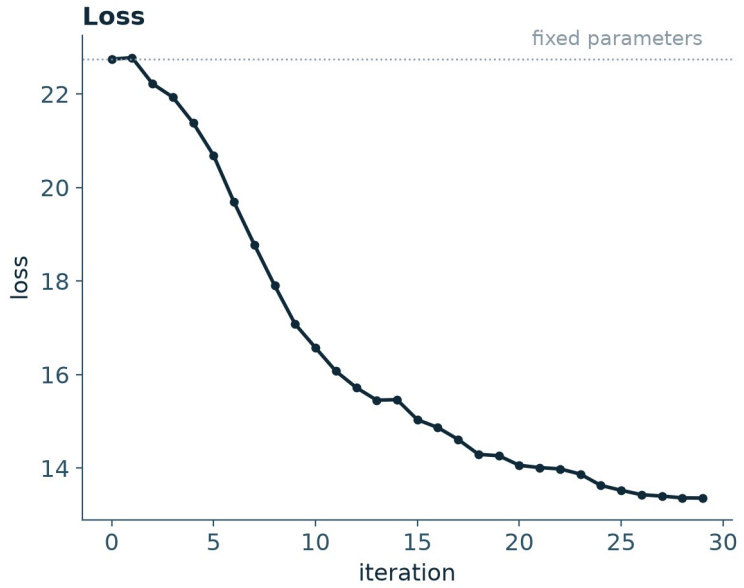
P, T_s, DDF, A, τ_ref (slide 5), with no control favoured.

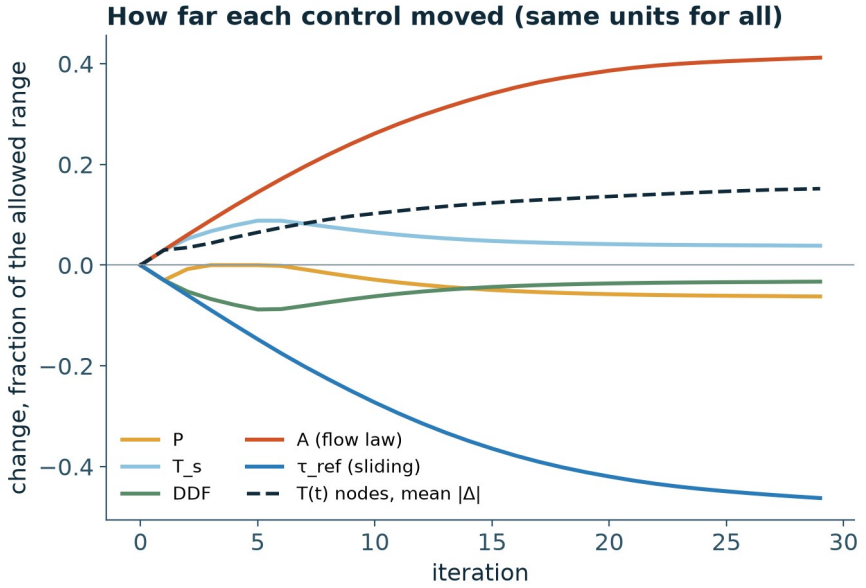
control	start	allowed range
T(t), 24 nodes	100 m inversion	literature envelope
P	1.8 m/yr	1.0 – 1.8 m/yr
T_s	9.4 °C	7.9 – 10.9 °C
DDF	10 mm/K/d	8 – 12 mm/K/d
A (flow law)	78 MPa^{-3}/yr	52 – 117
τ_ref (sliding)	0.215 MPa	0.10 – 0.25 MPa

Equal weight

- Every control is a fraction of its range, 0 … 1
- One Adam step size for all, no prior on the parameters
- T_present re-fitted to the geodetic balance at each step

Ice flow does the work, not the temperature





−41 %

loss 22.7 → 13.4 in 30 iterations

0.215 → 0.146

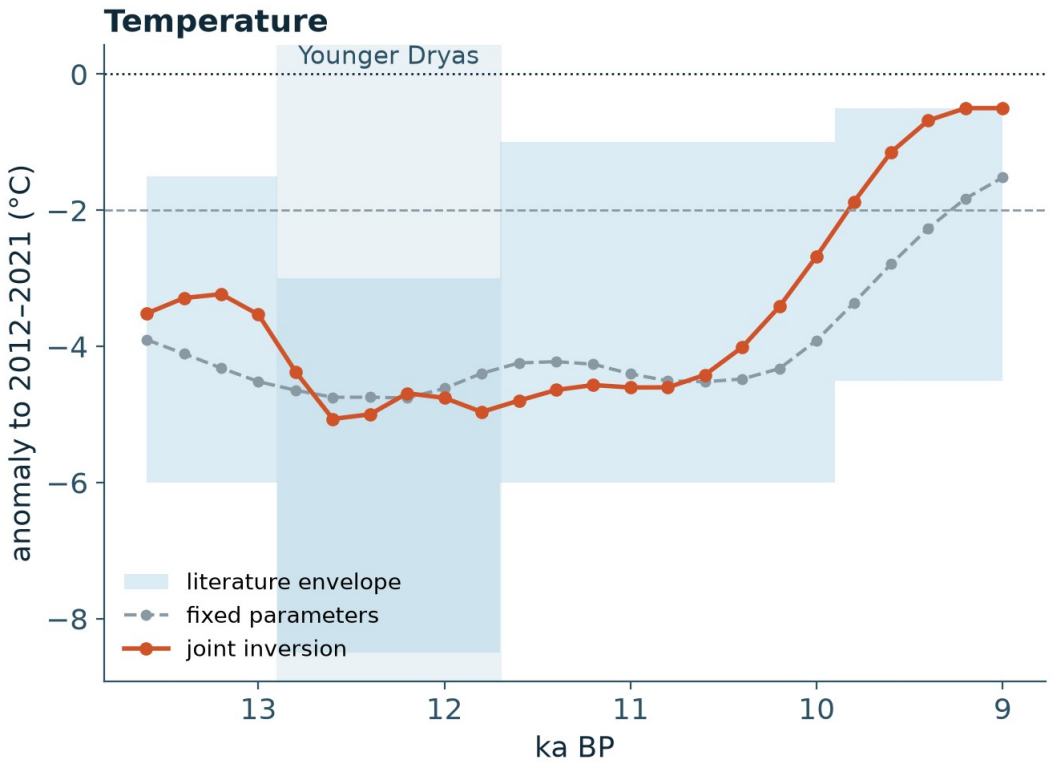
τ_ref (MPa): more sliding

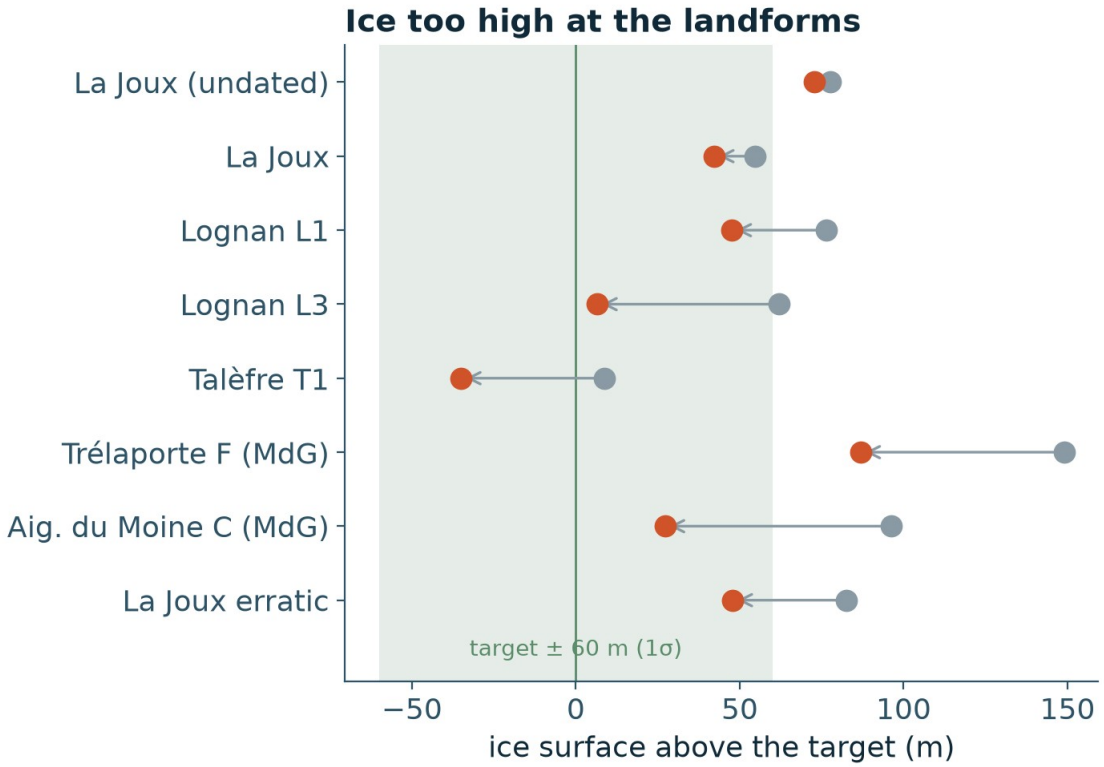
78 → 105

A (MPa⁻³/yr): softer ice

control	start	joint best	moved (fraction of range)
τ_ref (sliding)	0.215 MPa	**0.146 MPa**	**0.46**
A (flow law)	78 MPa⁻³/yr	**105 MPa⁻³/yr**	**0.41**
T(t), YD mean anomaly	−4.65 °C	−4.81 °C	0.15 (all nodes, mean)
P	1.80 m/yr	1.75 m/yr	0.06
T_s	9.40 °C	9.52 °C	0.04
DDF	10.0 mm/K/d	9.87 mm/K/d	0.03

Same Younger Dryas, thinner glacier





−4.81 °C

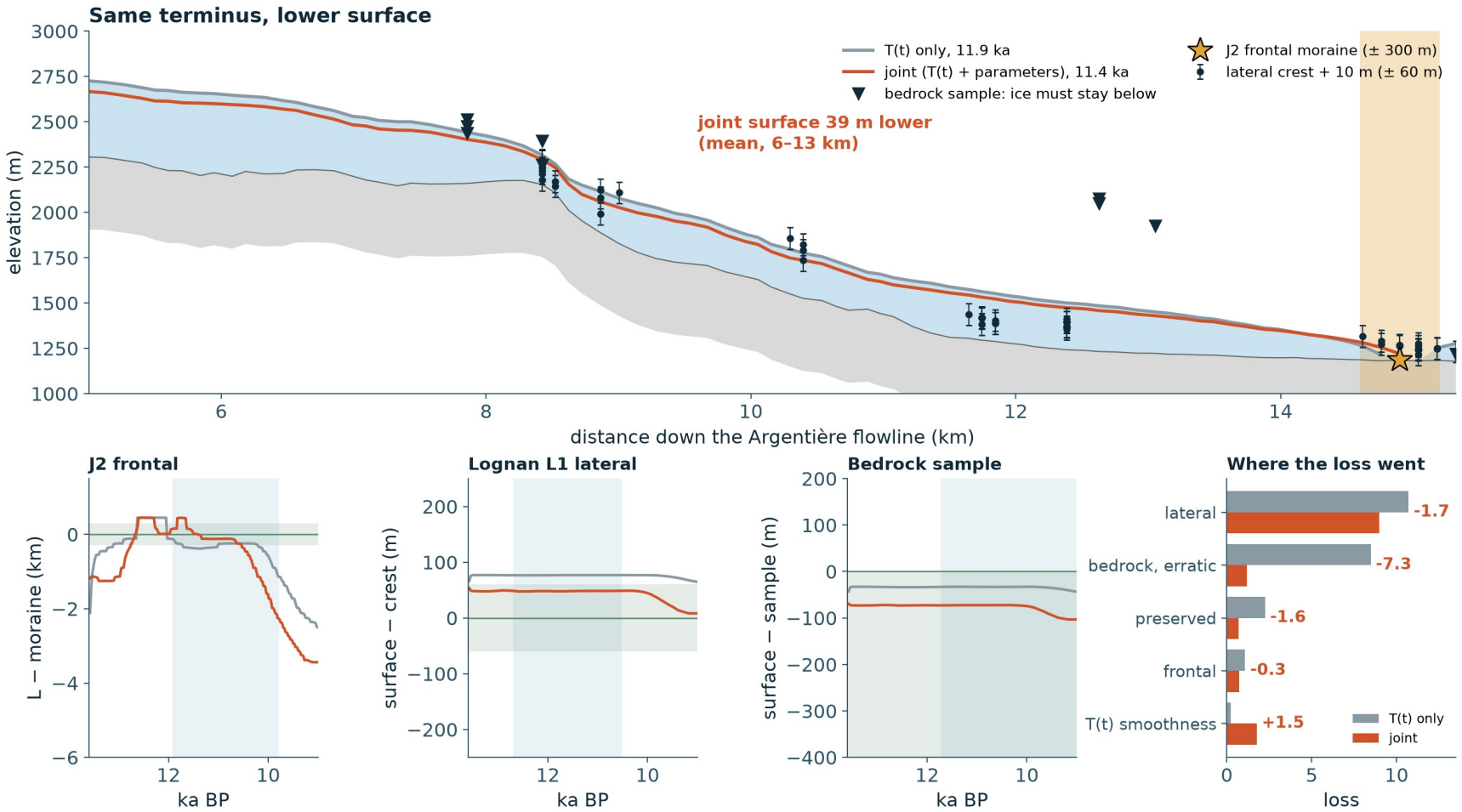
YD vs 2012–2021 (fixed parameters −4.65)

4 → 3

bedrock samples under > 10 m of ice

Not yet converged: τ_ref, A still drifting. 9.2–9.0 ka on the warm bound; La Joux still 40–75 m too high

How the fit improved: same length, thinner ice



39 m thinner

Same J2 terminus, lower surface along the whole valley.

Faster flow

More sliding (τ_ref ↓) and softer ice (A ↑) carry the ice further with less thickness.

Bedrock −7.3

Most of the gain: samples no longer under ice. Laterals −1.7.

Cost +1.5

T(t) a bit rougher (smoothness term).